

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
AUGSD/AGSRD
Course Handout

In addition to the part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : MPBA G505

Course Title : Statistics & Basic Econometrics

Instructor-in-charge : Udayan Chanda [Email: udayanchanda@pilani.bits-pilani.ac.in chamber no.- 6068W]

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1. Course description:

Probability and Probability Distributions, Characteristics of Probability Distributions, Some Important Probability Distributions, statistical inference, estimation and hypothesis testing; ANOVA, MANOVA, MANCOVA, the linear regression model (two-variable model, multiple regression, functional forms, dummy variables); regression analysis in practice (model selection criteria and tests, multicollinearity, heteroskedasticity, autocorrelation), simultaneous-equations models. Statistical package 'R' will be used in this course.

2. Scope and Objective:

This course serves as an introductory exploration of statistical analysis and fundamental econometrics, wherein the principles of economic theory, mathematics, and statistical inference are employed to examine business phenomena. The curriculum encompasses a review of probability distributions, statistical inference, estimation, and hypothesis testing. It also covers analysis of variance (ANOVA), analysis of covariance (ANCOVA), and introduces multiple regression analysis, which remains the most widely utilized statistical technique within the field of econometrics. The latter part of the course addresses various practical considerations related to linear regression models, including the use of dummy variables, alternative functional forms, and the implications of violating classical regression assumptions. Students will engage in the study and application of regression analysis to diverse data sets, thereby gaining familiarity with core concepts such as the estimation of economic parameters, forecasting business outcomes, and conducting statistical inference.

A weekly two-hour laboratory session will be conducted, during which students will learn to explore and analyze different types of economic data using R programming. This component aims to develop proficiency in advanced statistical and econometric techniques through practical applications in R.

3. Course learning outcomes:

By the end of this course, students should be able to:

CLO1: Apply statistical and econometric models to estimate relevant business parameters, predict economic outcomes, and test hypotheses using quantitative data.

CLO2: Understand the fundamental assumptions underlying the classical linear regression model and identify and address potential violations, such as autocorrelation and heteroscedasticity.

CLO3: Develop and sustain a foundational knowledge of econometrics that will serve as a basis for further study in business analytics and advanced statistical methodologies.

4. Text books and references

- TB1: Anderson, D. R., Sweeney, D. J., Williams, T. A., Camm, J. D., & Cochran, J. J. (2019). Statistics for business & economics. Cengage Learning.
- TB2: Gujarati, D. N., Porter, D. C., & Gunasekar, S. (2012). Basic econometrics. Tata McGraw-Hill Education.
- RB1: Heiss, F. (2016). Using R for introductory econometrics. Florian Heiss.
- RB2: Verzani, J. (2018). Using R for introductory statistics. CRC press.
- RB3: Hatak, N. R. (2010). Principles of Econometrics: An Introduction (using R). SAGE Publications India.
- RB4: Madnani, G. M. K. (2015). Introduction to Econometrics: Principles and Applications. Oxford and IBH Publishing.

5. Course Plan:

A. Lecture (Theory)

Wk. No.	Module No.	Description	Ref.	Learning Outcome
1	Module1	Introduction to Probability: Random Experiments, Basic Relationships of Probability, Conditional Probability, Bayes' Theorem	Chapter 4 of T1 + Class Notes	Understand the fundamentals of probability, apply conditional probability, and solve problems using Bayes' theorem.
2	Module2	Discrete Probability Distributions: Random Variables, Discrete Probability Distributions, Expected Value, Variance, Bivariate Distributions, Covariance	Chapter 5 of T1 + Class Notes	Develop and analyze discrete probability models and compute expectations, variances, and covariance.
3-4	Module3	Continuous Probability Distributions: Uniform, Normal, Normal Approximation to Binomial, Exponential Distribution	Chapter 6 of T1 + Class Notes	Apply continuous probability distributions to real-world problems and interpret probability models.
5	Module4	Sampling and Sampling Distributions: Sampling Distributions, Central Limit Theorem, Point Estimation, Confidence Intervals for Means and Proportions	Chapters 7 & 8 of T1 + Class Notes	Understand sampling theory, derive sampling distributions, and estimate population parameters.
6	Module5	Hypothesis Testing – I: Null and Alternative Hypotheses, Type I & II Errors, Tests for Population Mean	Chapter 9 of T1 + Class Notes	Formulate statistical hypotheses and conduct hypothesis tests for population means.
7	Module 5	Hypothesis Testing – II: Tests for Population Proportion, Chi-square Test of Independence, Goodness-of-Fit Test	Chapters 9 & 12 of T1 + Class Notes	Perform hypothesis tests for proportions and apply chi-square tests for categorical data.
8	Module 6	Analysis of Variance (ANOVA): One-way ANOVA	Chapter 13 of T1 + Class Notes	Compare multiple population means using one-way ANOVA and interpret results.

9	Module 7	Introduction to Econometrics: Definition, Scope, Goals, Methodology, Types of Econometrics, Regression vs Correlation and Causation	Chapter 1 of T2 + Class Notes	Explain the scope and objectives of econometrics and distinguish regression from correlation and causation.
10	Module 8	Two-Variable Regression Analysis – I: Population Regression Function, Linear Regression Model, Stochastic Disturbance, OLS Method	Chapters 2 & 3 of T2 + Class Notes	Develop simple linear regression models and estimate parameters using Ordinary Least Squares (OLS).
11	Module 8	Two-Variable Regression Analysis – II: Classical Linear Regression Assumptions, Maximum Likelihood Estimation, Goodness of Fit (R^2)	Chapter 4 of T2, Chapter 5 of R4 + Class Notes	Evaluate regression assumptions, assess model fit, and interpret regression results.
12	Module 9	Multiple Regression Analysis – I: Multiple Regression Equation, Partial Regression Coefficients, Multiple Correlation	Chapter 7 of T2 + Class Notes	Estimate and interpret multiple regression models involving several explanatory variables.
13	Module 9	Multiple Regression Analysis – II: Specification Bias, R^2 and Adjusted R^2	Chapter 7 of T2, Chapter 7 of R4 + Class Notes	Evaluate model adequacy and understand the implications of omitted variable bias and adjusted goodness of fit.
14	Module 10	Dummy Variable Regression Models: Dummy Variables, ANOVA Models, ANCOVA Models	Chapter 9 of T2 + Class Notes	Apply dummy variable techniques to model qualitative factors in regression analysis.
15	Module 11	Violations of Regression Assumptions – I: Autocorrelation and Heteroscedasticity	Chapters 10 & 11 of T2, Chapter 9 of R4 + Class Notes	Diagnose and interpret violations of regression assumptions and their impact on estimation.
16	Module 11	Violations of Regression Assumptions – II: Multicollinearity and Remedial Measures	Chapter 12 of T2, Chapter 10 of R4 + Class Notes	Identify multicollinearity, assess its consequences, and apply appropriate remedial measures in regression analysis.

B. Practical – Lab

Experiment Plan

Week	Module	Experiment Title	Learning Outcome
1-2	Probability Distributions	Generate and Visualize discrete and continuous distributions using the statistical environment	- To understand how to explore various distributions and their properties. - To demonstrate the CDF and PDF of Popular distributions to describe populations: exponential, Weibull, lognormal distribution.
3-4	Sampling distributions & Statistical inference	- Summary Statistics - Correlation and Covariance - Confidence Intervals - Significance Test - Goodness of Fit	- To understand the basics of statistical inference using computation - How to find a confidence interval for a parameter To understand the distribution of the values in a table. - How to find the test statistic based on how well the actual counts for each category fit the expected counts.
5-6	Simple Linear Regression Model & Analysis of variation	- The simple linear regression model - One-way ANOVA - Two-way ANOVA - An analysis of covariance (ANCOVA)	- How to use 'lm' to find the regression estimates - How to use 'predict' for predicting the response - Assessing the linear model for the mean, residuals, Influential points, and Prediction intervals - How to use 'oneway.test'/'aov'/'lm' to perform ANOVA - To understand Kruskal–Wallis test for equivalence of means - How to perform an ANCOVA in R - Fitting the two-way ANOVA model in R
7-8	Multiple Linear Regression Model	Estimation and Hypothesis Testing, and Interpretation	- To understand how to interpret the coefficients of the multiple linear regression model. - To understand how to get variance and covariance matrix for the estimated coefficients. - To understand how to test hypotheses about single coefficients in a multiple regression model.
9	Violation of Classical Assumptions of Regression -1	Consequences, Detection, and Remedies of Heteroskedasticity and Autocorrelation	- To understand how heteroskedasticity and autocorrelation influence the OLS estimator. - To understand how to compute the generalized least squares estimates under the influence of heteroskedasticity and autocorrelation
10	Violation of Classical Assumptions of Regression -2	Consequences, Detection, and Remedies of Collinearity	- Identify collinearity in regression. - Understand the effect of collinearity on regression models.

11	Simultaneous Equations Models	• Estimation of Simultaneous equations with 2SLS • Joint Estimation of System Equations	To understand how to use 'system fit in R to analyze system equations
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4. Evaluation Scheme:

	Component	Duration	Marks	Weightage (%)	Date & Time	Remarks
A. Theory	Mid Semester	90 minutes	80	26.67%		OB
	Quiz		40	13.33	To be announced in class	CB
	Comprehensive	180 minutes	100	33.33%		CB
	Total		220	73.33%		
B. Practical	Project		40	13.33	To be announced in class	OB
	Comprehensive Exam	To be announced in class	40	13.33		CB
	Total		80	26.67%		
Total (A+B)			300	100%		

5. Chamber Consultation Hours: To be announced in class

6. Notice: Notices concerning this course will be displayed on Management Department NB.